Game Engine Architectures 01

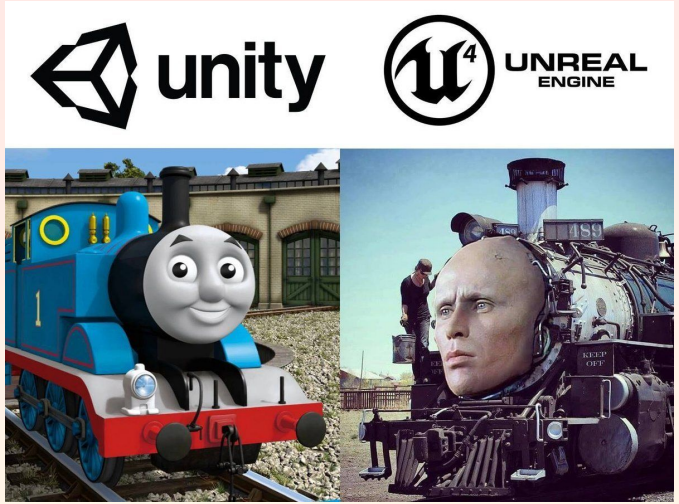
Introduction, General && Map Building

What is this class about?

I've been told to teach you Unreal Engine

> so that's what I'm gonna do!

We'll cover basics of working in UE, programming with Blueprints and C++, and a few basic concepts in the evaluation of games



Overview

Assignments && Project

- **Goal:** Creation of your own small video game in UE
- 1 submission at the end
- 2 Milestone submissions in-between
- The game content is up to you, but **keep the scope reasonably small**!
- Submissions will be through moodle, but the project should be uploaded to Github

Mandatory Project Content

The project should include at least the following:

- 2 scenes (1 can be the main menu)
- a 3D environment
- Player Input
- fitting Lighting (not the standard one)
- Physics of some sort
- simple UI
- some Sounds/Music

Everything else is up to you!

Overview Topics

01 - General, Map Building

- Navigating inside UE
- Fab Store
- Building a Map
  - Landscaping
  - Painting the Terrain
  - Lighting && Fog

02 - Blueprints, Basics

- Working with Blueprints
- Player Controller && Input
- Raycasts
- Physics && Collisions

Overview Topics

03 - UI, Events, Best Practices

- Game Interfaces
  - Dialogue UI etc.
- Events and Messages
- Best Practices in UE programming

04 - C++, State Machine

- C++ Programming Basics
- State Machine
- Nav Mesh && Agents
- Animation Controller

Overview Topics

05 - Eval, Memory Management

- Evaluation of Games (Theory)
- Memory Management
- Performance Optimization
- Making a Build

06 - Topic of Choice

- For example:
  - Audio System
  - Dynamic Post Pro Effects
  - more Map Building things
  - detailed AI Agents
  - Data Management
  - Saving && Loading, etc...
- Please think of one that would interest you or that you need for your game!

Overview Topics

>> Result from Lessons

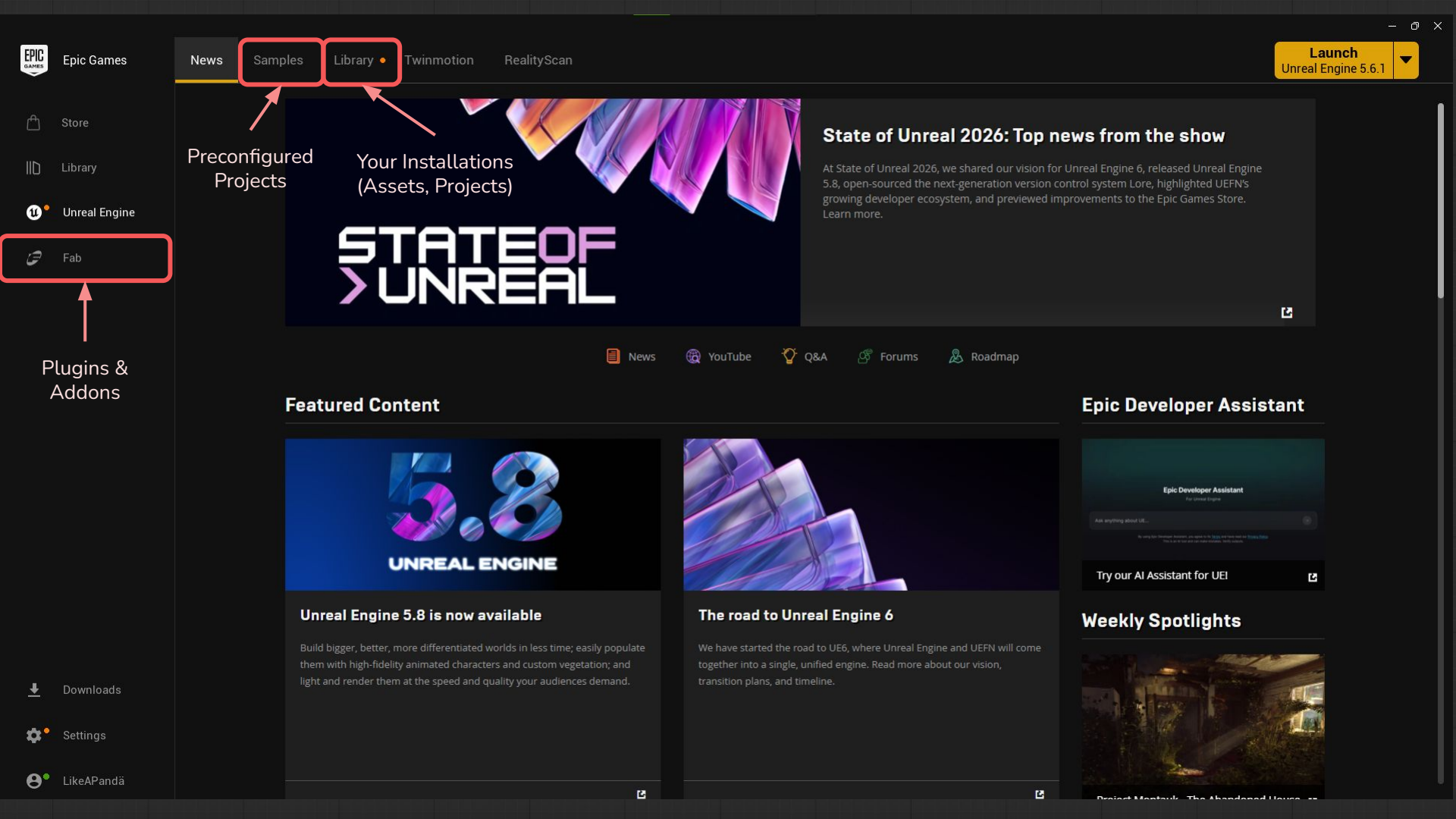
- very simple recreation of

  “Slender: The Eight Pages”

- Working on one game during class
- Will be built upon with every lesson



Unreal Engine Basics

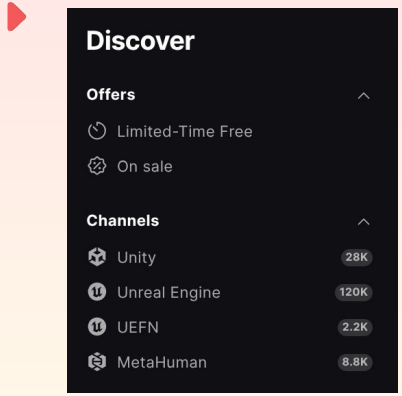

Epic Games
News
Samples
Library
Twinmotion
RealityScan
Launch
Unreal Engine 5.6.1
Store
Library
Unreal Engine
Fab
Preconfigured Projects
Your Installations (Assets, Projects)
Plugins & Addons
STATE OF UNREAL
State of Unreal 2026: Top news from the show
At State of Unreal 2026, we shared our vision for Unreal Engine 6, released Unreal Engine 5.8, open-sourced the next-generation version control system Lore, highlighted UEFN's growing developer ecosystem, and previewed improvements to the Epic Games Store. Learn more.
News
YouTube
Q&A
Forums
Roadmap
Featured Content
5.8
UNREAL ENGINE
Unreal Engine 5.8 is now available
Build bigger, better, more differentiated worlds in less time; easily populate them with high-fidelity animated characters and custom vegetation; and light and render them at the speed and quality your audiences demand.
The road to Unreal Engine 6
We have started the road to UE6, where Unreal Engine and UEFN will come together into a single, unified engine. Read more about our vision, transition plans, and timeline.
Epic Developer Assistant
Epic Developer Assistant
Try our AI Assistant for UEI
Weekly Spotlights
Downloads
Settings
LikeAPandä

Fab Store

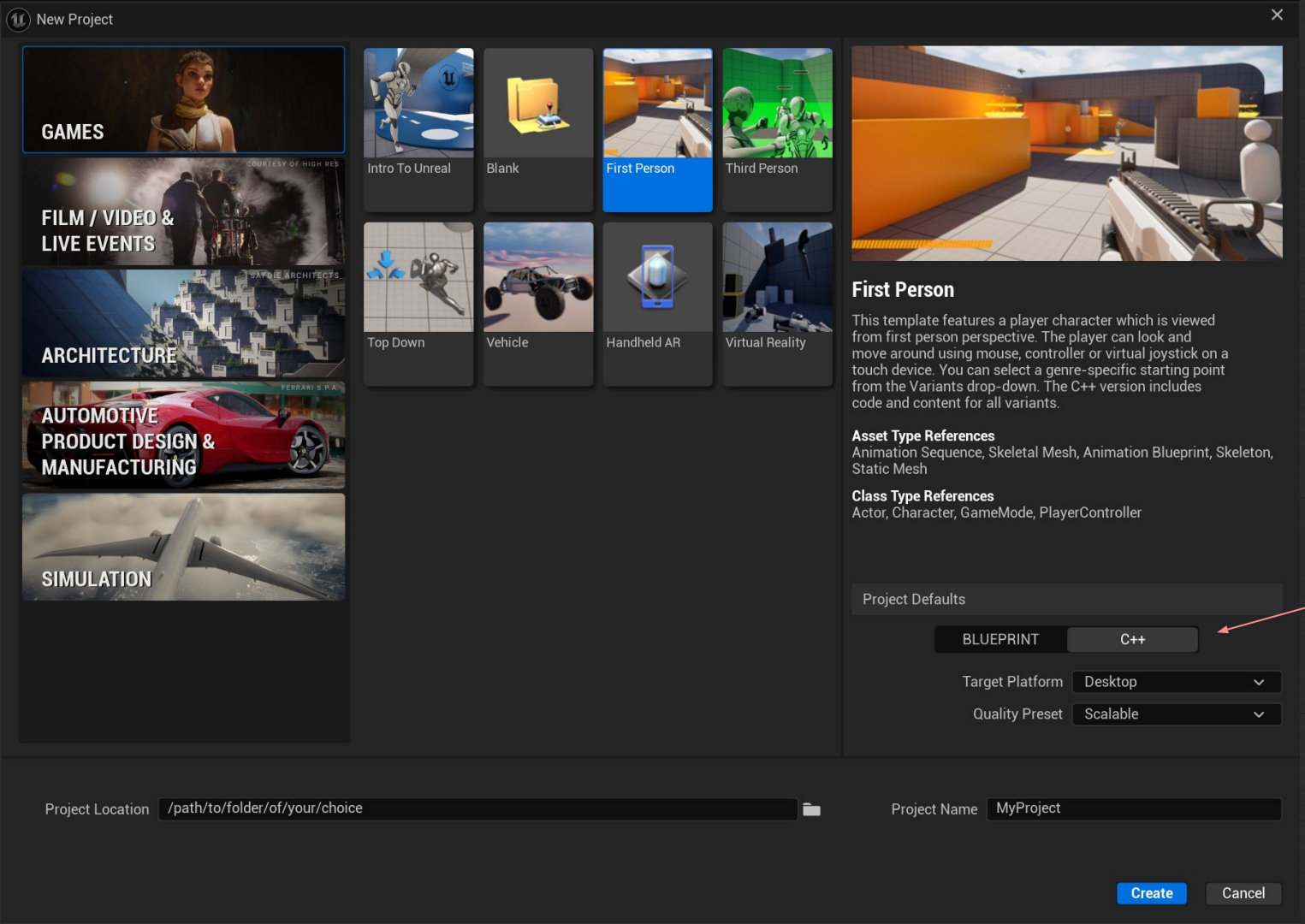
Fab can provide:

- 3D models
- textures/materials
- audio
- tools
- particle effects
- and more!

can be added directly from within the Engine



www.fab.com/



There's a lot of templates to choose from! That can make starting with your game a lot easier

make sure to choose "C++" here if you plan on using C++ at some point, otherwise you'll have to include the package manually later

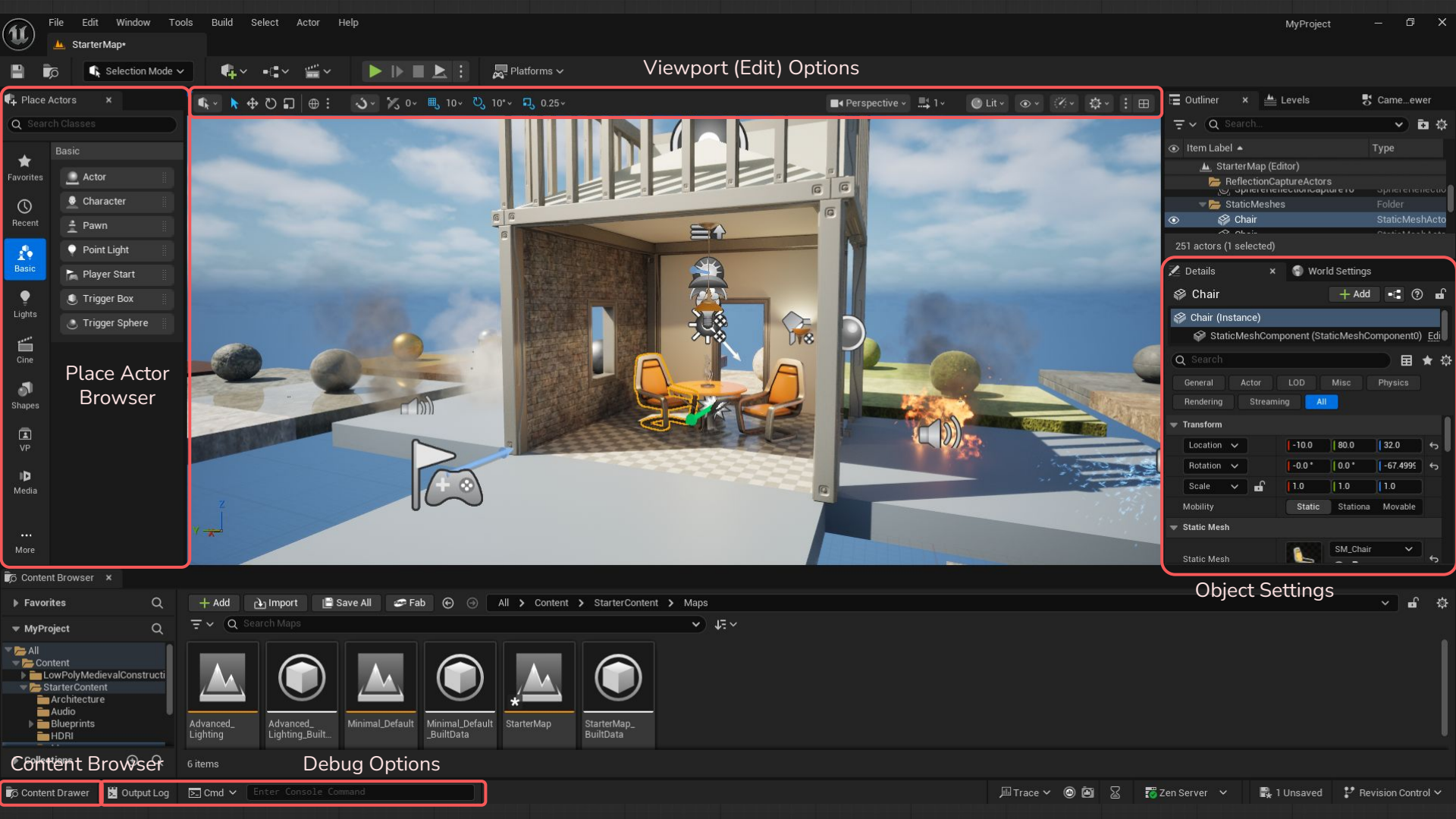
Unreal Engine Game Setup

- Use the correct UE version
  > we use *5.8.3* for this lesson
- You can choose any template and decide to work with C++ or not
  > we work with *C++* and use the “*First Person*” Template
- Please name your project clearly
- Store it somewhere you can find it again
- Connect Github/Version Control from the start
- Avoid spaces, umlauts, and special characters in project/folder names

UE is huge, so make sure you have enough space on your SSD!
(at least 100GB)




Program Settings
Viewport Options
File
Edit
Window
Tools
Build
Select
Actor
Help
MyProject
StarterMap*
Project & Level Options
Selection Mode
Platforms
Level Settings
Place Actors
Search Classes
Basic
Actor
Character
Pawn
Point Light
Player Start
Trigger Box
Trigger Sphere
Favorites
Recent
Basic
Lights
Cine
Shapes
VP
Media
More
Perspective
Lit
Outliner
Levels
Item Label
Type
StarterMap (Editor)
Lighting
SkyAtmosphere
SkyLight
SM_SkySphere
251 actors
Details
World Settings
Select an object to view details.
Content Browser
Favorites
MyProject
All
Content
LowPolyMedievalConstructi
StarterContent
Architecture
Audio
Blueprints
HDRI
Collections
Add
Import
Save All
Fab
All
Content
StarterContent
Maps
Search Maps
Advanced_ Lighting
Advanced_ Lighting_Built...
Minimal_Default
Minimal_Default _BuiltData
StarterMap
StarterMap_ BuiltData
6 items
Debug & Logging
Content Drawer
Output Log
Cmd
Enter Console Command
Trace
Zen Server
1 Unsaved
Revision Control


File Edit Window Tools Build Select Actor Help
MyProject
StarterMap*
Selection Mode
Platforms
Viewport (Edit) Options
Place Actors
Search Classes
Basic
Favorites
Recent
Basic
Lights
Cine
Shapes
VP
Media
More
Actor
Character
Pawn
Point Light
Player Start
Trigger Box
Trigger Sphere
Place Actor Browser
Perspective
Lit
Outliner
Levels
Item Label
Type
StarterMap (Editor)
ReflectionCaptureActors
StaticMeshes
Folder
Chair
StaticMeshActo
251 actors (1 selected)
Details
World Settings
Chair
Add
Chair (Instance)
StaticMeshComponent (StaticMeshComponent0)
General
Actor
LOD
Misc
Physics
Rendering
Streaming
All
Transform
Location
Rotation
Scale
Mobility
Static
Stationa
Movable
Static Mesh
SM_Chair
Object Settings
Content Browser
Favorites
MyProject
All
Content
LowPolyMedievalConstructi
StarterContent
Architecture
Audio
Blueprints
HDRI
Add
Import
Save All
Fab
All
Content
StarterContent
Maps
Search Maps
Advanced_ Lighting
Advanced_ Lighting_Built...
Minimal_Default
Minimal_Default _BuiltData
StarterMap
StarterMap_ BuiltData
6 items
Content Browser
Debug Options
Content Drawer
Output Log
Cmd
Enter Console Command
Trace
Zen Server
1 Unsaved
Revision Control

Hot Keys #1

- **Q/W/E/R** to choose transform option
- **Hold RMB + WASD** to move around in the scene (hold RMB + Mouse Wheel to change speed)
- **Spacebar** to cycle through transform options
- **To change Pivot:** under *Modelling Mode* > *XForm* > *Edit Pivot*
- **Hold Shift + drag** actor to move along with it
- **Hold Left Alt + drag** actor to copy it (**+ Shift** to move along with it)
- Right click actor > **Pilot Actor**
- **End key** to snap actors to ground
- Select actor + Select Menu > Material > **With Same Material** (or similar settings)
- **Click actor + H** to hide it; **CTRL + H** to unhide all
- **Click on actor + CTRL + B** to find it in the Content Drawer
- **Click on static mesh actor + CTRL + E** to open static mesh editor
- Isolate: click RMB on actor > Visibility > **Show Only Selected**; bring back with **CTRL + H**

Hot Keys #2

- Bind “Paste Here” command to specific key (in Project Settings); then copy actor, click on another surface, paste actor here with chosen key
- **Ctrl + Spacebar** to toggle content drawer
- **G** to toggle Game View
- **F11** to toggle full screen
- **Click on actor + F** to focus it
- **Hold RMB + C** to zoom in
- **Hold RMB + Z** to zoom out

    > Permanently change FOV in Perspective Dropdown
- View Transform Bookmarks: Look at something, then press **CTRL + number button 0-9**; move away in scene, then press number button 0-9 to move to that bookmarked view again
- **hold CTRL + L** to control directional light
- **CTRL + P** to open asset panel

Resources #1

Megascans



- choose from a variety of assets directly in the Engine (**Fab**)
- high quality meshes for **free use**
- also often offer different materials

https://www.youtube.com/watch?v=fFy-muKWmQ8

Nanite



- simply restructure meshes with one click for **higher performance** and **less storage usage**

https://www.youtube.com/watch?v=MRwcaBd7aGM

Resources #2

Lumen



- build **realistic world lighting**
- uses a fully dynamic indirect lighting pipeline

https://www.youtube.com/watch?v=ivFl0FbTJ7M

https://www.youtube.com/watch?v=Dc1PPYl2uxA

Chaos Physics

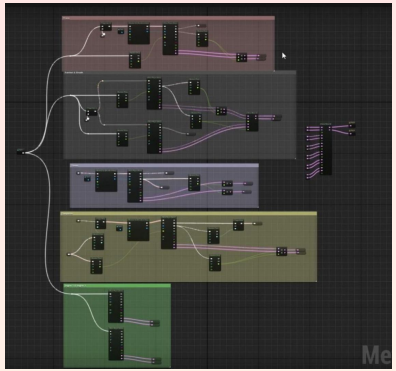


- create simulations of **physically correctly breaking** objects

https://www.youtube.com/watch?v=CeI7dSGc_j0

Resources #3

Metasounds



- Audio Studio for creating own dynamic sounds inside the Engine

https://www.youtube.com/watch?v=fF5Ev-bxTdY

https://www.youtube.com/watch?v=uckpxH42jQ0

Metahumans && Animator



- use 3D-Scan or create own character with Metahuman Creator
- animating through Motion Capture

https://www.youtube.com/watch?v=7kBrQDvhZXg&t=91s

Map Building in UE

Resources

DataSmith Exporter



- Connection between 3D or CAD software to simply synchronise 3D assets

https://www.unrealengine.com/en-US/plugins

Water and Landmass



- Plugins to quickly create waterbodies and landmass and customise them

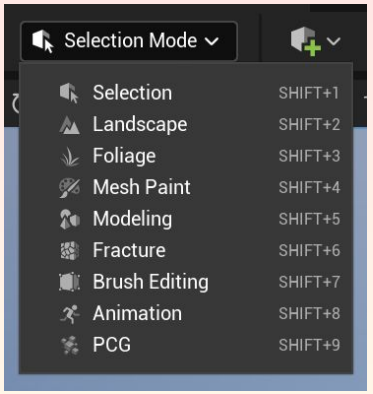
https://www.youtube.com/watch?v=6_

e.g.

https://www.fab.com/listings/6602874e-ef24-48c9-9055-a7ac07384696

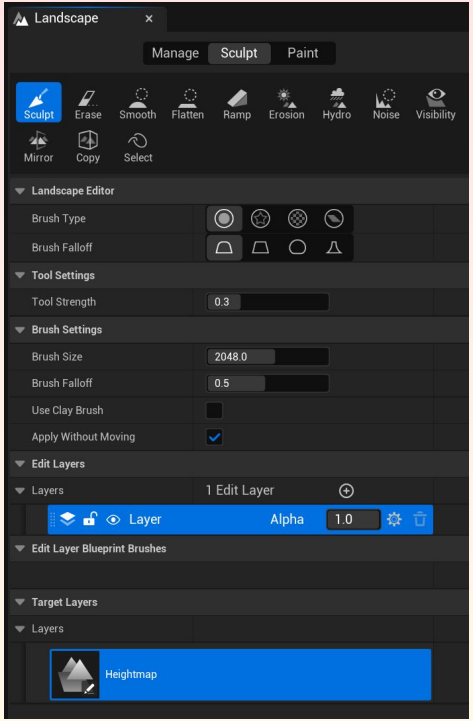
Different Modes

- switch between modes for specific tasks
  - > Landscaping
  - > Modeling
  - > Animation
  - > ...



Landscaping

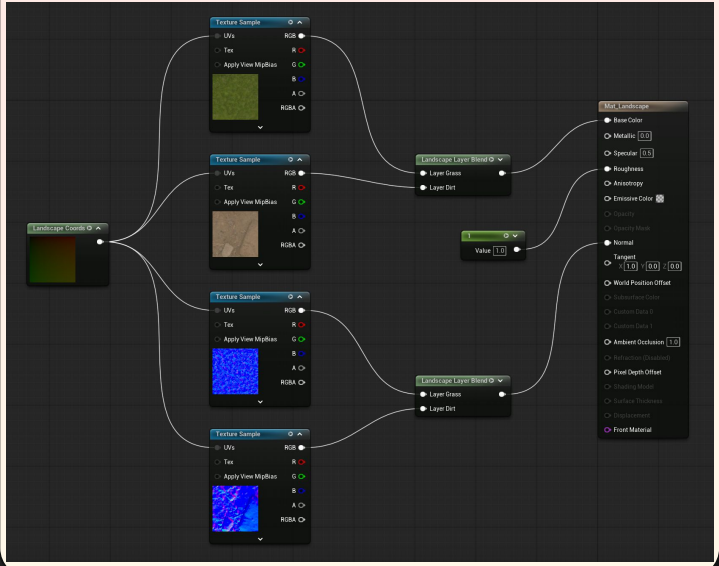
- go into the landscape mode to modify your terrain
- if there's none yet, it will automatically try to create one
  > change the settings as you need and click create
- (otherwise go to the "Manage" tab)
- there's many different sculpting tools at your beck and call



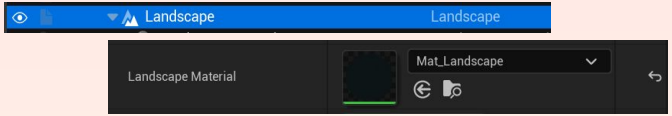
Painting the Terrain



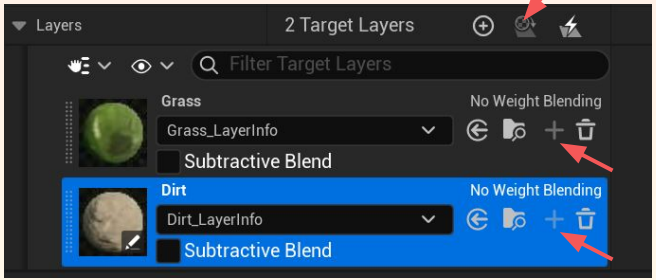
- Create a new Material and blend all your layers together



- Assign the material to the landscape...

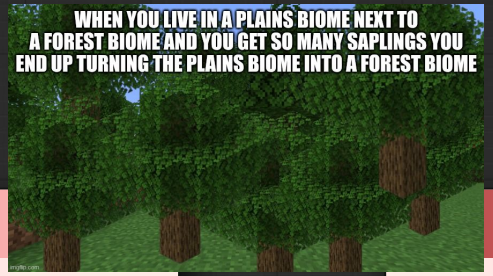


- ...create layers and layer infos from the material and start painting!

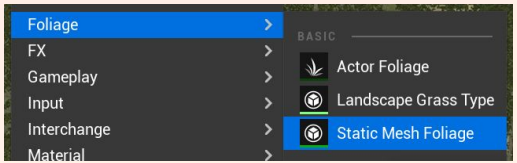


*hold Shift to erase!

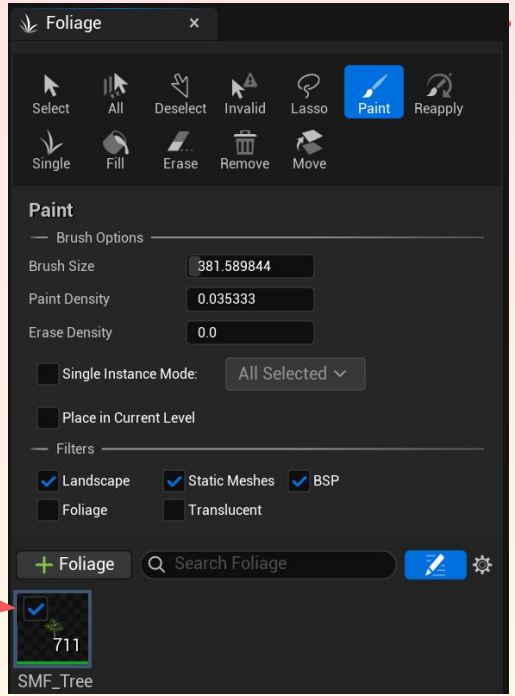
Placing Foliage



- to draw trees, grass etc., go into Foliage mode
- you need to first create a *Static Mesh Foliage* out of your mesh to be able to add it here



- for collisions, change **Instance Settings > Collision Presets** to *BlockAll* (and add a collider to your mesh)



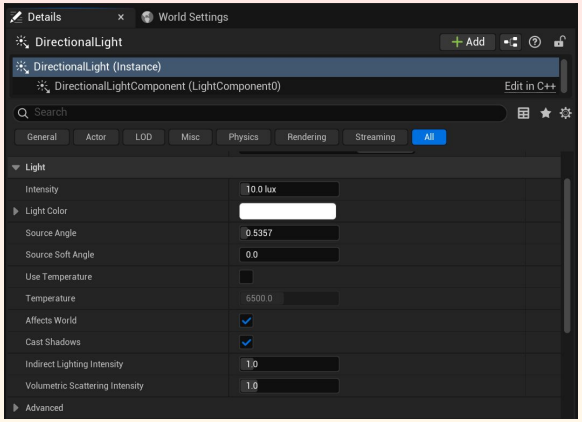
you need to check the foliage you want to draw with!

*hold Shift to erase!

Lighting #1

our basic scene lighting consists of 3 actors:

- **Directional Light** (the light that actually illuminates the scene)
- **Sky Atmosphere** (simulates sky and light scattering)
- **Sky Light** (matches lighting / reflections to appearance of sky)



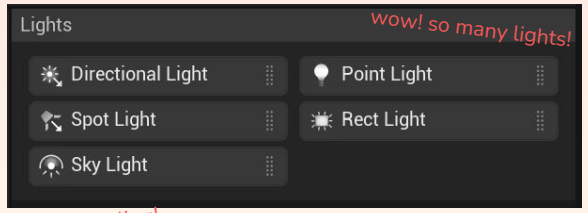
Hot tip

Hold Ctrl + L and move the mouse to change the lighting

Lighting #2

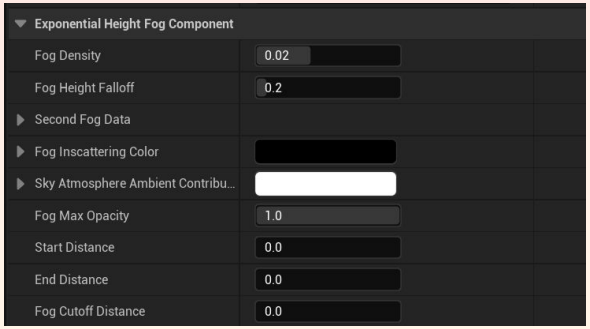
Three types of mobility

- **Static:** baked, cannot be changed while in game. Very performant!
- **Stationary:** partially baked, shadows and light bounces are static, colour and intensity can be changed
- **Movable:** light can be freely moved and changed in-game



Fog - Exponential Height Fog

- we can easily add some fog to set the mood
- UE has an own actor that can be added to the scene to enable fog
- change the color, density, start and end distance...



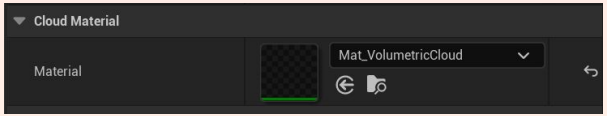
Place through...

Env. Light Mixer *or*
Visual Effects > Exponential Height Fog

Clouds - Volumetric Cloud

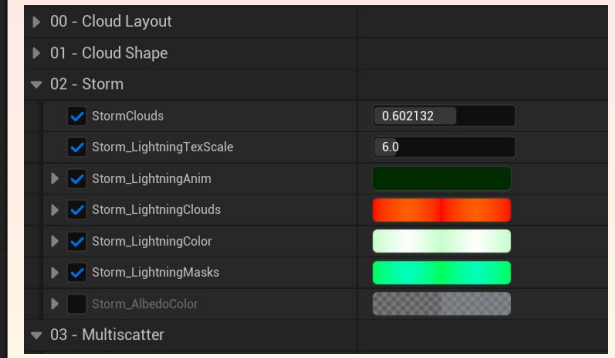
- and we can also change up the clouds and add a storm





- reveal this material in the browser, copy it to a local folder and edit it



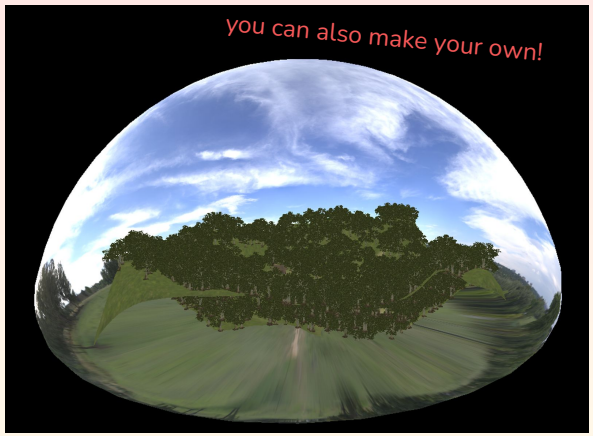


Place through...

don't forget to assign your new material!

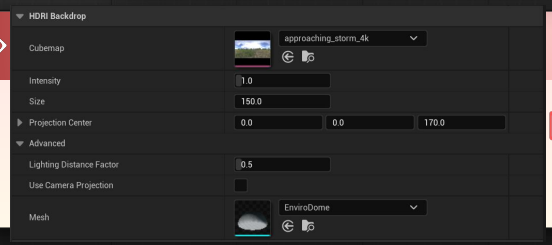
HDRI - High Dynamic Range Image

HDRIs are used to imitate a Horizon or/and a Sky to the player;
it encapsulates the scene



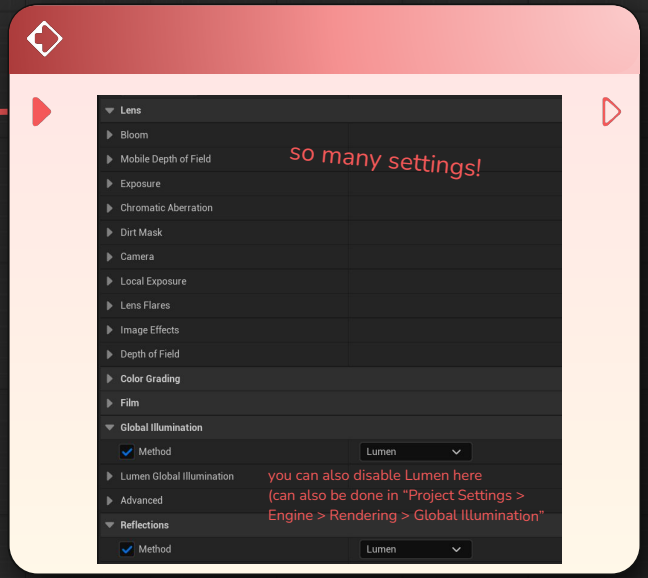
How to add an HDRI?

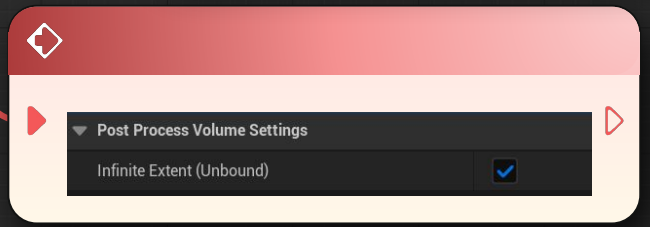
- go to the Plugins Window
- enable “HDRI Backdrop”
- download a free HDRI Image
- open the Texture in Unreal and change “Mip Gen Settings” to “NoMipMaps” (we want no compression!)
- place an HDRI Backdrop in the Level and
- change the “Cubemap” and “Mesh”



Post Processing

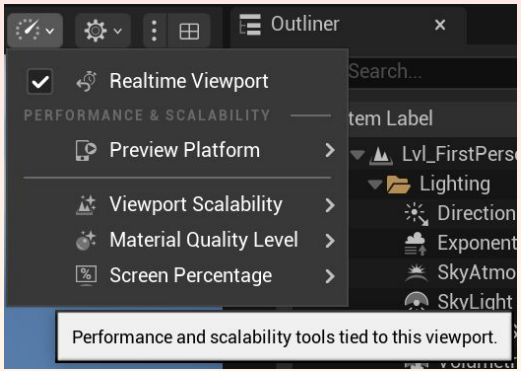
- Post Processing Volumes work the same way in UE as in other Engines
- it's a camera-level image polish
- it affects the whole image, not only specific objects
- PP can affect the whole scene (*global*) or just a specific area (*local*)
- in UE, to make the PP Volume global, check the setting *"Infinite Extent (Unbound)"*





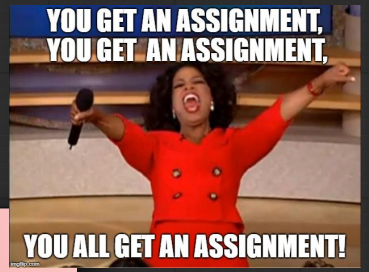
If your fps drop in edit mode...

- UE has performance and scalability tools:
- Viewport Scalability
  > turn everything to Medium
- Material Quality Level
  > Medium/Low are enough
- Screen Percentage
  > “Custom Override” & decrease
  > this helps the most



you can turn on FPS display at:
***Eye Symbol* > Viewport Stats > FPS**

Milestone Submission #1



Goal

Have a detailed game concept you can use as a guideline for the project

Keep the scope reasonably small!

Minimum Requirements

- at least 2 scenes (1 can be the main menu)
- a 3D environment
- Player Input
- fitting Lighting (not the standard one)
- Physics of some sort
- simple UI
- some Sounds/Music

What to submit

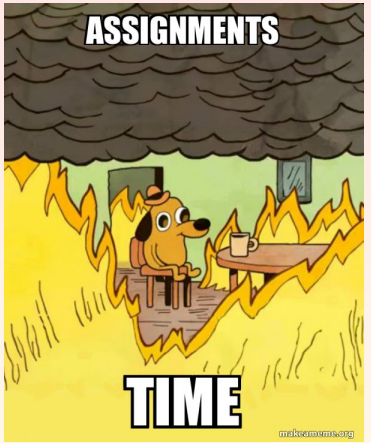
- 1-2 A4 pages, *.pdf* format
- describe your idea and add sketches / reference images, keeping in mind the requirements
- AI-generated texts are not allowed and will lead to deducted points

Due: 3rd of November

Milestone Submission #2

Goal

Have the 3D environment for your game project done



What to submit

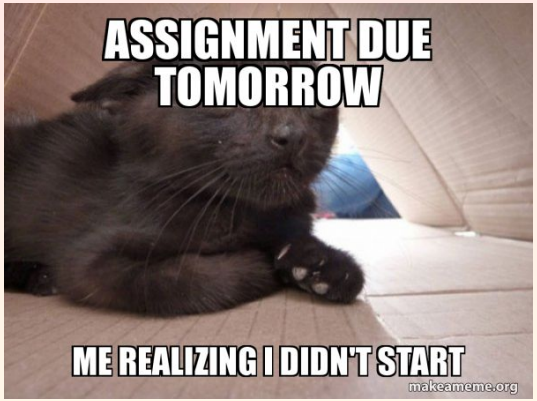
- *.zip* folder containing 5-10 screenshots of your 3D environment
- make sure to show different perspectives!

Due: 15th of December

Project Submission

Goal

✨ a finished videogame ✨



What to submit

.zip folder containing:
- a build of your game (windows or linux)
- .txt file with a link to your Github repo and any info you want to let me know

This is a **group project**, so you will receive just **one grade per team**! Should there be any issues, then please talk to me.

Due: 2nd of February

Version Control – Best Practices

- Most importantly: **Communicate**
- Avoid working on the same file at the same time
- Use Branches (for every feature, every team member, ...). Branches make merging a lot easier! Often merging can resolve conflicts automatically.
- Commit and Push little and very often.
- Commits are basically like backup saves. You can always go back to a previous commit!
- If you feel like the next thing you do could break something, then commit.
- You can also commit without pushing!
- Use clear commit subjects (and descriptions if deemed necessary). Often you can trace future issues through commits in the past > this makes it easier to check how e.g. code looked like back then.

Unreal Engine:

- Use the .gitignore file! Github has a template for UE
- **Binary files don't merge well!** Avoid working in the same binary file
  - Blueprints are also binary files!
- If Git LFS is necessary: only use for selected files and not the whole project!
  - Track binary files with LFS, not small code/text
  - Only use for files > 100 MB

